Diffusion Language Models: Adaptation from Autoregressive Models, Continual Pre-training, and Non-Autoregressive Text Generation

Generated survey — QUAL-SG (Bao et al., 2025) RAG-based Survey Generation task, implemented for this project. References [1]-[48] were retrieved and ranked by the QUAL-SG retrieval pipeline (`scripts/retrieve.py`, `scripts/arxiv_fetch.py`, `scripts/rerank.py`, `scripts/merge_finalize.py`); full metadata, abstracts, and per-criterion ranking scores for each are in `data/candidates_topK.json`.

1. Introduction

Large language models built on the autoregressive (AR), left-to-right decoding paradigm have driven most recent progress in natural language generation, but this paradigm imposes a strict generation order that limits parallelism, bidirectional planning, and easy revision of already-generated content. Diffusion models, which generate data by learning to reverse a gradual noising process, offer an alternative: they can attend to context in both directions, revise tokens after they are first produced, and generate many positions in parallel. Adapting this idea to text has produced two related but distinct lines of work that this survey brings together. The first is **text diffusion modeling** itself — building continuous or discrete diffusion processes over token sequences [1, 42, 29, 30, 31, 40]. The second is the broader family of **non-autoregressive (NAR) text generation** methods, which relax the strict left-to-right constraint using a variety of mechanisms beyond diffusion, from fully parallel decoding [39] to semi-autoregressive block generation [24, 31] and insertion-based generation [38]. A third thread, **continual pre-training and model adaptation** [32, 36, 20, 27], has emerged as a practical way to reuse the enormous investment already made in pretrained AR language models rather than training new architectures — including diffusion language models — from scratch. Recent large-scale diffusion language models increasingly combine all three threads: they take a pretrained AR checkpoint and continually adapt it into a diffusion model [14], echoing exactly the strategy this survey's anchor topic is concerned with. This survey organizes the literature along these three axes, tracing the technical lineage from early NAR machine translation and continuous text diffusion through masked/discrete diffusion language models to the current generation of AR-to-diffusion adaptation efforts, and closes with a review of very recent (2025-2026) work that has appeared since the most prominent AR-to-diffusion adaptation papers were published.

2. Background: From Diffusion Models to Non-Autoregressive Text Generation

Diffusion probabilistic models learn a forward process that gradually corrupts data with noise and a reverse process that denoises it step by step, and they have been highly successful for continuous data such as images. Applying the same idea to text is complicated by the discrete nature of language: tokens cannot be perturbed with continuous Gaussian noise in the same way pixels can. Two broad strategies have been used to bridge this gap. The first embeds discrete tokens into a continuous space and applies standard Gaussian diffusion there, as in Diffusion-LM [1] and DiffuSeq [42]. The second defines the forward noising process directly over the discrete token space — e.g., by randomly replacing tokens with a special `[MASK]` symbol or resampling them from a categorical distribution — giving rise to *discrete*

or *masked* diffusion models [40, 31, 30]. Both strategies inherit from the wider non-autoregressive generation literature, which predates modern text diffusion by several years and originally targeted machine translation: Gu et al. [39] introduced fully parallel NAR translation to cut inference latency at some cost to quality, and subsequent work explored semi-autoregressive relaxations [24], iterative refinement [23], and explicit position modeling [33] to close the resulting quality gap. Diffusion models can be viewed as a particularly structured and theoretically grounded instance of this broader NAR family, since the sequence of denoising steps provides a principled, learnable iterative-refinement schedule rather than a hand-designed one.

3. Text Diffusion Models

3.1 Continuous Diffusion for Text

Diffusion-LM [1] is among the first models to adapt continuous denoising diffusion to text generation, embedding tokens into a continuous space and using gradient-based methods on the diffusion process to enable fine-grained, plug-and-play controllable generation (e.g., over syntactic structure) that autoregressive LMs struggle to support without retraining. DiffuSeq [42] extends continuous diffusion to conditional sequence-to-sequence generation, showing that a single diffusion model can support diverse Seq2Seq tasks (paraphrasing, generation, simplification) without a separate cross-attention encoder-decoder architecture. GENIE [29] proposes a large-scale pretraining framework for continuous diffusion LMs, introducing a continuous paragraph denoising objective so the diffusion decoder learns to reconstruct coherent multi-sentence text from noise, rather than being trained only on short spans. TGM-DLM [17] applies a similar continuous diffusion LM approach outside natural language, generating SMILES strings for text-guided molecule design, illustrating that continuous text diffusion transfers to other symbolic sequence domains once framed as sequence generation from noise.

3.2 Discrete and Masked Diffusion for Text

Because text is inherently discrete, a second line of work defines the diffusion process directly over the token vocabulary instead of a continuous embedding space. Reparameterized discrete diffusion models [12] derive an equivalent, more flexible formulation of sampling from discrete diffusion processes, improving both training stability and decoding quality. A cheaper and better diffusion LM with soft-masked noise [48] identifies that naively applying continuous-style Gaussian corruption is a poor match for discrete text and proposes a soft-masking corruption scheme designed specifically for language. Diffusion-NAT [22] connects discrete diffusion explicitly back to the NAR generation literature, using a self-prompting discrete diffusion process built on top of pretrained language models for non-autoregressive text-to-text generation, and explicitly frames discrete diffusion as a way to resolve the incompatibility between earlier continuous diffusion approaches and standard pretrained LM tooling. Simple and effective masked diffusion language models (MDLM) [30] show that a carefully engineered, simplified masked-diffusion training recipe with a Rao-Blackwellized objective substantially narrows the previously reported performance gap between diffusion and autoregressive language models — a key result that later scaling efforts (§3.4, §6) build on directly.

3.3 Initializing Diffusion Language Models from Pretrained (Masked) LMs

A recurring idea is to avoid training a diffusion LM from scratch by initializing it from an existing pretrained masked or autoregressive LM. DiffusionBERT [40] combines a discrete diffusion process with a BERT

initialization, arguing that BERT's own masked-denoising pretraining objective is already closely related to the diffusion training objective, so BERT's weights provide a strong starting point for a generative discrete diffusion model. SSD-LM [31] instead builds a semi-autoregressive diffusion LM: it iteratively generates blocks of text with a simplex-based diffusion process, combining the flexible-length generation and modular, plug-and-play controllability of diffusion with block-wise autoregressive-style decoding for efficiency. This initialize-from-an-existing-LM strategy is the direct conceptual predecessor of the more recent trend (§3.4, §4.3) of adapting *large-scale autoregressive* LLMs — rather than smaller masked LMs like BERT — into diffusion models.

3.4 Scaling Masked Diffusion Language Models (2024-2026)

Building on the MDLM recipe [30], recent work has pushed masked/discrete diffusion LMs toward much larger scale and toward matching autoregressive LLMs on general instruction-following and reasoning tasks rather than only on language modeling perplexity. TESS 2 [14] is a particularly direct precedent for AR-to-diffusion adaptation: it builds a general instruction-following diffusion LM by first *continuing pretraining* a strong existing AR model with a diffusion loss and only then performing instruction tuning, and the authors report that both this adaptation step and the choice of base AR model are crucial to final quality — closely paralleling the anchor paper's own adaptation recipe. Dream-Coder 7B [9] applies the same adapt-from-pretrained-AR-model philosophy specifically to code generation, and shows that a diffusion LM can learn to *adaptively choose* its own decoding order (sketch-first for complex algorithms, left-to-right for simple completions), an emergent any-order capability that autoregressive decoders cannot express by construction. Beyond capability scaling, a growing body of 2025-2026 work analyzes and improves the diffusion LM training and inference process itself: Svete and Sabharwal [4] give a theoretical characterization of what reasoning problems masked diffusion models can and cannot efficiently solve, connecting them to chain-of-thought and padded looped-transformer computational models; Ringel et al. [16] identify that parallel unmasking degrades quality when the tokens chosen for simultaneous decoding are statistically dependent, and propose a lightweight dependency predictor to guide which positions are safe to decode in parallel; and DiSPO [3] introduces a plug-in reinforcement-learning credit-assignment layer that optimizes intermediate masked-filling decisions rather than only the final output, addressing the coarse credit assignment that comes from training a multi-step iterative refinement process with only a terminal reward.

4. Adapting and Continually Pre-training Language Models

4.1 Continual / Domain-Adaptive Pre-training of Autoregressive LMs

Training a large language model from scratch is extremely resource-intensive, which has motivated a large body of work on *continuing* to pretrain an already-trained model instead. Ke et al. [32] study continual domain-adaptive pre-training explicitly, proposing a method to continually adapt an LM across a sequence of unlabeled domain corpora while mitigating forgetting of earlier domains. Gupta et al. [36] study a practical ingredient of this process — how to "re-warm" the learning rate schedule when resuming pretraining on new data — showing that naive continuation can cause large, avoidable performance regressions on previously learned distributions. Xie et al. [27] and Fujii et al. [20] apply continual pre-training to build domain- and language-specialized models (a financial-domain LLM and a Japanese-capable LLM built by extending Llama 2's vocabulary and continuing pretraining on Japanese web text, respectively), demonstrating that continual pre-training is now a standard, practical recipe for

specializing an existing open AR LLM rather than training a new one.

4.2 Architecture Transfer: From Transformers to Alternative Architectures

A related strand of adaptation work targets not a new *domain* but a new *architecture*. Mamba [28] proposes a selective state-space model as a subquadratic-time alternative to the Transformer's attention mechanism, and subsequent work (outside this survey's core retrieved set, but widely cited alongside it) has shown that such alternative architectures can themselves be obtained by transferring the weights of an existing pretrained Transformer rather than training from random initialization — the same "reuse an existing pretrained model to bootstrap a differently-structured model" logic that underlies AR-to-diffusion adaptation. This makes architecture-transfer methods a close methodological cousin of AR-to-diffusion adaptation: both treat a pretrained AR Transformer as a reusable source of learned linguistic knowledge that can be redirected into a model with a fundamentally different computational structure (a different attention/state mechanism in one case, a bidirectional non-causal diffusion denoiser in the other).

4.3 Adapting Autoregressive Models into Diffusion Models

Combining §3.4 and §4.1-4.2, the most direct precedents for adapting large pretrained AR LLMs into diffusion LMs are TESS 2 [14] and Dream-Coder 7B [9], both of which explicitly start from a pretrained autoregressive checkpoint and continue training it with a diffusion objective before further tuning. Planner and Executor [21] takes an alternative, non-weight-sharing approach to combining the two paradigms: rather than converting an AR model into a diffusion model, it studies hybrid architectures in which a discrete diffusion model and an autoregressive model collaborate at inference time (e.g., one plans while the other executes), reporting complementary benefits from the two decoding styles. Together, these results indicate that the field has converged on continual pretraining with a diffusion objective as the standard recipe for building large diffusion LMs efficiently, while collaborative AR+diffusion inference remains a less-explored complementary direction.

5. Non-Autoregressive Text Generation

5.1 Foundational Non-Autoregressive Machine Translation

Gu et al. [39] introduced non-autoregressive neural machine translation, removing the sequential dependency between output tokens to obtain an order-of-magnitude reduction in decoding latency, at a cost in translation quality that a large subsequent literature has worked to close. Glancing Transformer [23] narrows this gap with a curriculum-style "glancing" sampling strategy during training, and Gu and Kong [43] systematically catalog and combine several such tricks to produce fully non-autoregressive models competitive with iterative or semi-autoregressive alternatives. PNAT [33] argues that explicitly modeling the *positions* of generated words as a latent variable — rather than assuming a fixed, monotonic alignment — is essential to closing the quality gap, and Ran et al. [47] similarly inject explicit reordering information into the NAR decoding process to mitigate the "multimodality problem" in which multiple valid translations correspond to the same source sentence.

5.2 Semi-Autoregressive and Position-Aware Models

Between fully parallel and fully sequential generation lies a spectrum of semi-autoregressive methods that keep some local sequential structure while parallelizing across blocks. Wang et al. [24] propose the semi-autoregressive Transformer, which preserves autoregressive dependencies globally across blocks

but generates multiple tokens per block in parallel; SmBoP [25] applies an analogous bottom-up, semi-autoregressive strategy to semantic parsing, decoding top-k sub-trees at each height of the target abstract syntax tree rather than a single top-down path; and Deng and Rush [15] propose cascaded decoding with bounded-context Markov transformers, achieving sub-linear-time parallel generation while retaining most of the modeling benefits of autoregressive conditioning. SSD-LM [31] (§3.3) can be read as a diffusion-based instantiation of this same semi-autoregressive principle. Yang et al. [2] instead constrain the parallel decoding process directly with part-of-speech information, showing that structural linguistic constraints can guide which positions are safe to decode in parallel — a similar idea to the recent dependency-aware parallel decoding for diffusion LMs discussed in §3.4 [16].

5.3 Non-Autoregressive Generation Beyond Translation

Non-autoregressive decoding has been applied well beyond machine translation. Liu et al. [34] apply NAR decoding to unsupervised sentence summarization by first performing an edit-based search for a heuristic pseudo-ground-truth and then training an encoder-only NAR model on the resulting search output. Su et al. [44] show that a pretrained BERT encoder can serve as an effective backbone for NAR generation more broadly, addressing the classic NAR weaknesses of fixed output length and conditional-independence-induced repetition. The same parallel-decoding idea has also been adapted to non-text-generation-adjacent modalities and tasks sharing the same sequential-output structure: video captioning [37], scene-text recognition [46], non-autoregressive automatic speech recognition (Paraformer [45] and the comparative study of Higuchi et al. [41]), and even non-autoregressive diffusion-based time-series forecasting [7], underscoring that "generate a structured sequence in parallel rather than strictly left-to-right" is a general modeling pattern rather than a translation-specific trick. Huang et al. [35] provide a theoretical account of *why* NAR training is challenging in the first place, showing that the multimodality problem underlying many of these applied results follows from simplifying assumptions common to standard NAR training objectives.

5.4 Constrained and Insertion-Based Generation

A further branch of NAR-adjacent work targets *lexically constrained* generation, where the output must contain a given set of keywords. POINTER [38] generates text by progressively inserting new tokens between existing ones, expanding a template that already contains the required constraint words, rather than decoding purely left-to-right or fully in parallel. AutoTemplate [26] offers a simpler recipe for the same lexically-constrained setting by casting it as a template-filling task, trading some of POINTER's architectural complexity for implementation simplicity. Mansimov et al. [8] generalize the connection between undirected sequence models (such as BERT) and text generation more broadly, providing a unifying framework that foreshadows why masked/discrete diffusion LMs — which are themselves undirected, denoising models — are a natural fit for flexible-order and infilling-style generation (see also DDOT [13] in §3.4/§6, which targets flexible-length infilling specifically for discrete diffusion models).

6. Recent Developments Since DiffuLLaMA (2025-2026): Scaling, Reasoning, and Applications

Because the anchor paper for this survey (Gong et al., *Scaling Diffusion Language Models via Adaptation from Autoregressive Models*, ICLR 2025) was written in 2024, our recency-oriented retrieval surfaces a substantial body of directly relevant work published after it, which the anchor paper's own related-work section could not have cited. This growth reflects rapid consolidation around the "adapt a

pretrained AR model into a diffusion LM" recipe: TESS 2 [14] and Dream-Coder 7B [9] (§3.4) both scale this recipe to general instruction-following and code generation respectively. Several 2025-2026 papers dig into *why* and *how well* the resulting models work: Svete and Sabharwal [4] formally analyze the reasoning capability of masked diffusion LMs; Berrayana et al. [21] explore hybrid diffusion-plus-autoregressive collaboration as an alternative to full conversion; and Ringel et al. [16] and DiSPO [3] target specific weaknesses of the parallel-decoding process (dependency violations and coarse credit assignment, respectively) that only become apparent once diffusion LMs are trained at a scale where their generation quality is competitive enough to analyze closely. Zhang et al. [13] extend the infilling capability that motivated much of the original interest in diffusion LMs (§2, §5.4) to flexible, variable-length spans, and application-oriented papers such as text-attributed graph learning [19] and retrieval-augmented generation with diffusion LMs [5] show the paradigm being adopted as a component within larger systems rather than studied only in isolation. Taken together, this recent cluster indicates the field has moved from asking "can an AR model be adapted into a competitive diffusion LM at all" (the anchor paper's question) to "how do we scale, specialize, and analyze the diffusion LMs this adaptation recipe now reliably produces."

7. Open Challenges and Future Directions

Several open problems recur across the surveyed literature. First, the quality gap between non-autoregressive and autoregressive generation, while substantially narrowed by masked diffusion approaches [30], is not fully closed, and the theoretical work characterizing NAR/diffusion learning difficulty [35, 4] suggests some of the gap may be structural rather than purely a matter of scale or training recipe. Second, parallel decoding still faces a fundamental tension between speed and the conditional-independence assumption it relies on; dependency-aware decoding [16, 2] is a promising but still early direction for resolving this without sacrificing the parallelism that motivates NAR/diffusion approaches in the first place. Third, while AR-to- diffusion adaptation [14, 9] has proven more efficient than training diffusion LMs from scratch, it inherits open questions from the continual pre-training literature more broadly — how to avoid forgetting the base model's capabilities [36, 32] and how much adaptation data and compute are truly required — that have not yet been studied as thoroughly in the diffusion-adaptation setting as in the standard AR continual-pretraining setting. Finally, most of the retrieved diffusion-LM literature remains centered on English text and general - domain benchmarks; extending the adaptation recipe to multilingual and domain-specialized settings (in the way continual pre-training has already been extended, e.g. [20, 27]) remains comparatively unexplored for diffusion LMs specifically.

8. Conclusion

This survey traced three intertwined literatures — text diffusion models, continual pre-training/architecture adaptation, and non-autoregressive text generation — that converge on a now-dominant recipe for building large diffusion language models: start from a pretrained autoregressive LLM and continually adapt it with a diffusion training objective, rather than training a diffusion LM from scratch or initializing only from a small masked LM as earlier work did [40, 31]. Early non-autoregressive [39] and continuous/discrete text diffusion [1, 42, 12, 30] work established the modeling foundations; continual pre-training research [32, 36] established that reusing a pretrained model is both cheaper and often better than retraining; and a cluster of 2025-2026 papers [14, 9, 4, 3, 16] shows the adaptation recipe

now being scaled, analyzed, and specialized across code generation, reasoning, and retrieval-augmented settings. The remaining open challenges — closing the residual AR/NAR quality gap, decoding safely under token dependence, and avoiding catastrophic forgetting during adaptation — define the near-term research agenda for this area.

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